Dave's RFID Race Tracking System
Setup Manual
V 1.0

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Introduction

Welcome to the setup guide for Dave’s RFID Race Tracking System. This is a web-based system which uses Radio-Frequency Identification technology to track the laps made by participants in a race. The kind of race can be anything form go-carts, to drones and RC cars, to marathon racers or sprinters. Each form requires a vehicle or person to have an RFID tag on them.

Each RFID tag is registered into the system by a Race Attendant. This is the person who is controlling the back-end software. The Attendant may also view and change the status of a vehicle’s availability, update maintenance logs, view lap records, and most importantly, start and manage races.

In addition to this guide there will be a companion video guide to help you though the setup process and show the front and back end applications in action.

Dave’s RFID Race Tracking System

Requirements

The RFID Race Track System requires both hardware and software to run. You will need the following items:

- 1 Arduino Uno R3 Board
- 1 RC-522 RFID Reader
- Arduino IDE software
- Visual Studio Code
- A Web Browser
- MongoDB (Compass if running the Database locally, Atlas if online)
- Backend and Frontend Software downloaded off of GitHub

> Note: The RFID reader in this software is a placeholder for an UHF RFID Reader capable of reaching 5-10 Metres and scanning multiple tags at once. The RC-522 may also require you to solder the pins onto it.

> Waring: There is a known bug with the Mozilla Firefox browser which causes short perfomance lock-ups that get progressively worse over time. While this is the case, please use Google Chrome when using the software.

Back End Seup

RFID Reader Setup

Connect the RC-522 pins to the Uno R3 Board with the following diagram:

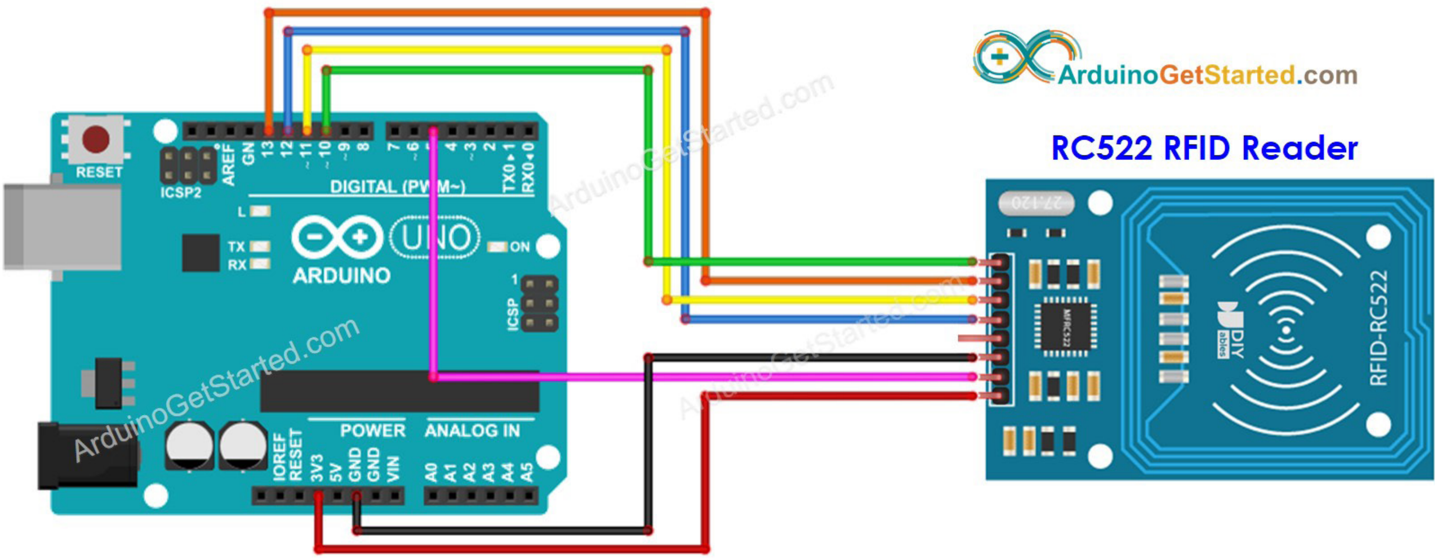

Image Courtesy of ArduinoGetStarted.com

Once the ports are appropriately connected, plug the Board into a usb port of your choice ofn the machine you’ll be using to run the Back End on.

Download the Back-End software. Inside, there is a folder named ArduinoCode, and the folder inside named RFID_Reader is the file you need to load onto the Arduino board named RFID_Reader.ino. Open Arduino IDE, choose the port your RFID reader is in and load the file onto it and save it. You may test the RFID reader by placing a tag onto the Reader and seeing the output on the Serial Monitor screen.

Please note the name of the port you select when loading onto the board. This will be necessary for setting up the backend application.

Note: The code for the RFID reader is from a tutorial from ArduinoGetStarted.com, modified to output the hex code into a string. You can find this tutorial at the following link:

https://arduinogetstarted.com/tutorials/arduino-rfid-nfc

Database Setup

You will now need to setup a local MongoDb database using Mongosh and MongoDb Compass. Be sure to create a user with readWrite priveliges for your server. Note the name and password of the user along with your database name and host

Open the .env file in the project and fill in the following details for each variable:

- DBUSER: The MongoDb username you chose
- DBPWD: The MongoDb password you chose for the user
- DBHOST: 127.0.0.1:27017 (this is the standard MongoDb address for locally hosted MongoDb Servers. If you selected a different port, use that port instead.)
- DB_NAME: the name of your server (we recommend race_tracker)
- SESSIONSECRET: whatever you like
- SALT: whatever you like so long as it's at least 16 characters long
- RFIDPORT: the name of the port the RFID reader is plugged into

You should now be ready to run the software. By default, the software runs on port 8888. If you change this, both the front-end and back-end must also be changed in the code.

In the terminal in Visual Studio Code, type the following command:

```
node app.js
```

Then, open localhost:8888 in a web browser.

> Waring: If the RFID is not plugged into the appropriate port on running app.js, the application will crash.

Login & Dashboard

On the login page, create a new user for the software. Type in a username and password and create a new user. Then, login to the Dashboard using those credetials.

Once on the Dashboard you will find the following sections:

- Vehicle Registry
- Races
- Utility

In the Vehicle Registry section, select the Add New Vehilce.

Adding A New Vehicle

To add a new vehicle, input the Vehicle Number, then scan the RFID tag you wish to use for that vehicle. The Hex code on the tag will be added to the form field for you. Select Add Vehicle and you’ll be brought back to the Dashboard where you should see a new vehicle listed in the Vehicle Registry.

Viewing & Updating A Vehicle

Click on View Vehicle on any registered vehicle you see. From here, you can see it’s status (active, idle, etc.). You can change details like the Vehicle Number, RFID hex code, or even delete the vehicle.

The Vehicle Maintenance section has a log of all repairs made by maintenance staff. To add a log, fill in the details in the Add Maintenance Log. Be sure to mark the vehicle as ‘idle’ if it will be available to use after the log is entered, otherwise mark it as “under maintenance’.

Running A Race

From the Dashboard, select Start New Race. From there, the Attendant gets the name and email address of each Racer participating in the race. Once the Add Racer button is pressed, the racer will be added to the race.

The Attendant must also select the number of laps being made for the race. Once that is done, the Attenant selects Start Race.

At this point, all racers should get into their vehicles/positions. Once the racers are ready, the Attendant his the Start Race button. A countdown will occur, and after 10 seconds, the race begins.

Scanning an RFID tag that belongs to a registered vehicle in the race will record and display that vehicle's lap time, racer name, vehicle number, etc. As more tags are scanned, those vehicles' data will be displayed.

Once all vehicles have made all of their laps, the Finish Race button appears. When the Attendant presses this button, the Finish Race page opens, displaying the pole positions of each Racer, and the option to either Start a New Race or return to the Dashboard.

Utility

Right now, the Utility section only has a Logout button. Use this to exit the Dashboard.

In the future, this will be the way to connect RFID readers, setup multiple accounts with different permissions, and so on.

Front End Seup

Introduction

The Front End of this applicaiton is a display for viewers and racers only. It takes information on the database via API calls and displays it as races are run.

Setup

Download the project from the GitHub repo. Open a new window in Visual Studio Code and open the project in it. Run the following command into the terminal:

```
npm run dev
```

Open a new browser window and go to localhost:5173

This should display a page with images and data that will change as new races are started, run, and concluded.